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Sec: 4

WEEK-1

1. Create a class Student with student ID, name and age. Create three student objects and display their details.

```
class Student {
  int studentId;
  String name;
  int age;

  void display() {
    System.out.println("Student ID: " + studentId);
    System.out.println("Name: " + name);
    System.out.println("Age: " + age);
    System.out.println();
  }

  public static void main(String[] args) {
    Student s1 = new Student();
    s1.studentId = 101;
    s1.name = "Pujith";
```

```
        s1.age = 20;

        Student s2 = new Student();
        s2.studentId = 102;
        s2.name = "Rahul";
        s2.age = 21;

        Student s3 = new Student();
        s3.studentId = 103;
        s3.name = "Kumar";
        s3.age = 19;

        s1.display();
        s2.display();
        s3.display();
    }
}
```

2. Create an Employee class with employee ID, name and salary. Initialize the values using a constructor and display employee details.

```
class Employee {
    int employeeId;
```

```
    String name;
    double salary;

    Employee(int id, String n, double s) {
        employeeId = id;
        name = n;
        salary = s;
    }

    void display() {
        System.out.println("Employee ID: " + employeeId);
        System.out.println("Name: " + name);
        System.out.println("Salary: " + salary);
    }

    public static void main(String[] args) {
        Employee e1 = new Employee(101, "Nitish", 30000);
        Employee e2 = new Employee(102, "Rahul", 40000);

        e1.display();
        e2.display();
    }
}
```

3. Create a BankAccount class with account number and balance. Implement deposit and withdrawal methods.

```
class BankAccount {
  int accountNumber;
  double balance;

  BankAccount(int accNo, double bal) {
    accountNumber = accNo;
    balance = bal;
  }

  void deposit(double amount) {
    balance = balance + amount;
    System.out.println("Deposited: " + amount);
  }

  void withdraw(double amount) {
    if (amount <= balance) {
      balance = balance - amount;
      System.out.println("Withdrawn: " + amount);
    } else {
      System.out.println("Insufficient Balance");
    }
  }
```

```
  void display() {
    System.out.println("Account Number: " + accountNumber);
    System.out.println("Balance: " + balance);
  }

  public static void main(String[] args) {
    BankAccount b1 = new BankAccount(12345, 5000);

    b1.display();
    b1.deposit(2000);
    b1.withdraw(1000);
    b1.display();
  }
}
```

4. Create a Rectangle class with length and width. Calculate and display its area and perimeter.

```
class Rectangle {
  double length;
  double width;

  Rectangle(double l, double w) {
```

```
        length = l;
        width = w;
    }

    void calculate() {
        double area = length * width;
        double perimeter = 2 * (length + width);

        System.out.println("Length: " + length);
        System.out.println("Width: " + width);
        System.out.println("Area: " + area);
        System.out.println("Perimeter: " + perimeter);
    }

    public static void main(String[] args) {
        Rectangle r1 = new Rectangle(10, 5);
        r1.calculate();
    }
}
```

5. Create a Book class with ISBN, title and price. Initialize the values using a parameterized constructor.

```
class Book {
```

```
    String ISBN;
    String title;
    double price;

    Book(String i, String t, double p) {
        ISBN = i;
        title = t;
        price = p;
    }

    void display() {
        System.out.println("ISBN: " + ISBN);
        System.out.println("Title: " + title);
        System.out.println("Price: " + price);
    }

    public static void main(String[] args) {
        Book b1 = new Book("9781234567890", "Java Programming", 450);
        Book b2 = new Book("9789876543210", "Python Programming", 500);

        b1.display();
        System.out.println();
        b2.display();
    }
}
```